AI/ML INTERNSHIP · DEVELOPERSHUB CORP · TASK 03

# Heart Disease *Prediction*

920 Patients · 13 Features · Binary Classification · Logistic Regression + Decision Tree

03

BY ABDUL SAMAD

| BEST ACCURACY | BEST ROC-AUC | PATIENTS TESTED | MISSING VALUES FIXED |
|---|---|---|---|
| 83.15% | 0.8949 | 184 | 1,759 |
| Logistic Regression winner | 89.5% sick/healthy separation | 20% held-out test set | Filled with column median |

MODEL EVALUATION · TEST SET

## Performance Comparison

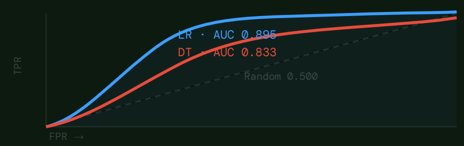
-- LOGISTIC REGRESSION 🏆

| | |
|---|---|
| Accuracy | 83.15% |
| ROC-AUC | 0.8949 |

-- DECISION TREE

| | |
|---|---|
| Accuracy | 77.17% |
| ROC-AUC | 0.8327 |

ROC CURVE




CONFUSION MATRIX · LOGISTIC REGRESSION

## What the Model Got Right

| | Predicted NO | Predicted YES |
|---|---|---|
| | ~72 TRUE NEG ✅ Correctly healthy | ~10 FALSE POS ⚠ False alarm |
| | ~21 FALSE NEG ❌ Missed! (worst) | ~81 TRUE POS ✅ Correctly caught |

| 920 | 736 | 184 | 13 |
|---|---|---|---|
| PATIENTS | TRAIN | TEST | FEATURES |

KEY FINDINGS

## What the Data Tells Us

01 **Asymptomatic chest pain = highest risk**
Pain that doesn't feel "normal" is paradoxically the most dangerous — the heart is in distress but not signaling it the typical way.

02 **Lower max heart rate = more disease**
Healthy hearts pump harder during exercise. A diseased heart can't keep up — its peak rate is measurably lower under stress tests.

03 **Blocked vessels = direct disease score**
More blocked arteries = more disease. Unlike other indirect signals, ca is a physical count of arterial damage.

04 **Age is NOT the top predictor**
A 35-year-old with poor heart function scores higher risk than a healthy 65-year-old. Tests beat age every time.

DATA CLEANING · 6 FIXES

## Before Training

- ✅ `num → binary target` — 0=healthy, 1–4=disease → 0 or 1
- ✅ `Dropped id + dataset` — Row number + hospital — useless
- ✅ `Text columns encoded` — sex, cp, restecg, slope, thal
- ✅ `Bool columns encoded` — fbs, exang: True→1 False→0
- ✅ `1,759 missing values` — Each filled with column median
- ✅ `thalch → thalach` — Column renamed for clarity

TECH STACK & SKILLS

## Libraries Used

- `scikit-learn` — Models, metrics, scaler
- `pandas · numpy` — Data cleaning & encoding
- `matplotlib · seaborn` — EDA + visualization

SKILLS

Classification · EDA · ROC-AUC · Confusion Matrix · Feature Importance · Encoding · Imputation · Stratified Split